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Global temporal observability of linear dynamic systems

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ABSTRACT

In this paper, we introduce a new concept termed global temporal observability for continuous and discrete linear dynamic systems and explore its connection with the classical notion of observability. It is shown that, as a concept, global temporal observability is a generalization of the classical observability. However, as a feature of a dynamic system, global temporal observability is embedded into classical observability. The necessary condition for global temporal observability is presented. Four linear systems were considered to test the proposed concept. Since observability is a binary test, our results matched the results of classical observability analysis when appropriate basis functions are utilized. The advantages and disadvantages of the proposed concept are discussed. The main advantage of global temporal observability is that it restores the state function for the entire time duration in a single step that requires matrix inversion. It is shown that global temporal observability connects state reconstruction, differential equations, and observability concepts.

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1. Introduction

The concept of observability for linear systems was introduced by (Kalman, 1963). Later it was extended to nonlinear systems using differential geometry (Hermann & Krener, 1977; Kou, Eliott, & Tarn, 1973) and differential algebra frameworks (Diop & Fliess, 1991a, 1991b). Necessary and sufficient conditions for observability of continuous and discrete linear time-varying systems are presented in Leiva and Sigmund (2003), Li, Mu, and Wang (2019), Reissig, Hartung, and Svaricek (2014), Silverman and Meadows (1967), Starkov (2002), Witczak, Puig, Rotondo, and Witczak (2017). The observability matrix was shown to be a Jacobian of the mapping from the state space to the differential embedding induced by the observation variables (Aguirre & Letellier, 2005; Letellier, Aguirre, & Maquet, 2005). The notion of observability was extended for network systems (Aguirre, Portes, & Letellier, 2018; Liu, Slotine, & Barabási, 2013), finite-state and heterogeneous multi-agent systems (Liu, Shen, Wu, & Su, 2021; Lozano, Spong, Guerrero, & Chopra, 2008; Sundaram & Hadjicostis, 2013), switched systems (Bejarano & Mera, 2021; Kang, Barbot, & Xu), and probabilistic systems (Chan & Shachter,

1992; Zhu, Jia, & e. Feng, 2023). The classical observability matrix gives a qualitative binary (“yes” or “no”) result for the observability analysis. To quantify the observability measure for linear and nonlinear systems, various observability indices were defined (Bianco-Martinez, Baptista, & Letellier, 2015; Friedland, 1975; Letellier & Aguirre, 2009; Letellier et al., 2005). Recently the research focus shifted to network theory-based observability analysis of dynamic systems, where graphs are used to represent information flow in dynamic models (Leitold, Vathy-Fogarassy, & Abonyi, 2017; Letellier, Sendiña Nadal, & Aguirre, 2018; Stigter, Joubert, & Molenaar, 2017; Wang, Gao, Gao, Deng, & Wang, 2014). Network-based analysis can analyze both static (structural view) and dynamic (transient) aspects of a system. One difficulty in using the observability concept is that the computation of the observability matrix for high-dimensional nonlinear systems is challenging due to unacceptable computation time and memory requirements (Stigter et al., 2017). Observability of linear systems from random measurements using compressive sensing principles was considered in Wakin, Sanandaji, and Vincent. The observability concept was extended to probability distributions, i.e. considering the observability of a distribution describing initial condition from the time evolution of output distribution, in Zeng and Allgöwer (2015), Zeng, Waldherr, Ebenbauer, and Allgöwer (2016). The functional observability property was introduced for linear and nonlinear systems in Jennings, Fernando, and Trinh (2011), Montanari, Freitas, Proverbio, and Gonçalves (2022).

The purpose of this paper is to propose a new concept of observability rather than suggest a solution to existing problems related to observability. The *observability* is a binary property

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(yes/no) of a system, that answers the question of whether the initial state of the system can be estimated from the input–output data (Zhou & Doyle, 1998). Similarly, the *reconstructability* is the binary property of a system, that answers the question of whether the final state of the system can be estimated from the input–output data (Kurek, 1987). *Detectability* of linear systems is related to the stability of the observer (Tse & Athans, 1973; Zhou & Doyle, 1998). For time-invariant linear systems, these two concepts are equivalent, while their difference becomes apparent for time-varying linear systems (Van Willigenburg & De Koning, 2008). Our main idea is to estimate, using inputs and output measurements, the state for all time duration under consideration instead of estimating the state at a given moment in time (initial or final time). In other words, we propose reconstructing the state vector for the whole interval of observation time in a single step involving matrix inversion. To accomplish this task using the classical observability analysis, two steps would be required: observability analysis of the initial condition and forward integration of the dynamic equations, or reconstructability analysis of the current condition and backward integration of the dynamic equations. The state of a system is often estimated using so-called observers, which are dynamic systems, whose solutions asymptotically approach the true state values (Bernard, Andrieu, & Astolfi, 2022; Tse & Athans, 1973; Zhou & Doyle, 1998). However, even the utilization of the observers (Luenberger, high-gain, sliding mode, extended), etc. to estimate the true state of the system for a finite time duration still requires integration of their extended dynamic equations, including the gain and covariance matrices (Nozaki, Sekiguchi, & Nonaka). The difference between our concept and the existing observability-related concepts is akin to the difference between differential and integral variational principles in classical mechanics. The former principles (that include the principles of virtual displacements, D'Alembert–Lagrange, Jourdain, Gauss) describe the motion at a specific moment in time. While the latter principles (that include the least-action principles of Hamilton, Lagrange, Jacobi) describe the motion for a finite period of time.

In summary, this paper proposes a new concept of the global temporal observability of linear dynamic systems. The main novelty of this work can be summarized as:

- The new concept of global temporal observability is introduced for linear dynamic systems.
- It is proven that if a linear system is globally temporally observable, then it is locally temporally (classically) observable.
- The necessary condition for global temporal observability is presented.

This work connects the following concepts: state reconstruction, differential equations, and observability of linear dynamic systems. The new concept can have impact on the theory of linear and nonlinear dynamic systems, especially for systems that do not satisfy the theorem of existence and uniqueness of solutions. Multiple simulations were performed to validate the proposed idea.

The paper is organized as follows. The Section 2 reviews classical notion of observability and introduces a new global temporal observability concept for continuous and discrete linear systems. Sections 3 and 4 describe the simulation performed and the results obtained. Section 5 discusses some aspects of the new concept, and the main results are summarized in Section 6.

2. Methods

2.1. Classical observability and reconstructability of linear continuous systems

We are concerned with the following linear continuous time-invariant system

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),\end{aligned}\quad (1)$$

where $\mathbf{x}(t) \in \mathbb{R}^{n_x}$, $\mathbf{u}(t) \in \mathbb{R}^{n_u}$, $\mathbf{y}(t) \in \mathbb{R}^{n_y}$ are, correspondingly, the state, input, and output vectors, and matrices $\mathbf{A} \in \mathbb{R}^{n_x \times n_x}$, $\mathbf{B} \in \mathbb{R}^{n_x \times n_u}$, $\mathbf{C} \in \mathbb{R}^{n_y \times n_x}$, $\mathbf{D} \in \mathbb{R}^{n_y \times n_u}$ have the dimensions compatible with the corresponding vectors. The initial condition is $\mathbf{x}_0 = \mathbf{x}(0)$. If no confusion arises, we omit the dependence of functions on time t . The observability is defined as the following²

Definition 1. A system (1) is called observable if, given an output $\mathbf{y}(t)$ and input $\mathbf{u}(t)$ signals over a finite time interval $0 \leq t \leq \tau$, it is possible to uniquely identify the initial state vector $\mathbf{x}(0)$.

Here τ is the length of the observation interval. Similarly, the reconstructability is defined as

Definition 2. A system (1) is called reconstructable if, given an output $\mathbf{y}(t)$ and input $\mathbf{u}(t)$ signals over a finite time interval $0 \leq t \leq \tau$, it is possible to uniquely identify the final state vector $\mathbf{x}(\tau)$.

The goal of the observability (reconstructability) analysis is to determine whether the state vector at a specific moment in time can be identified. Traditionally, the term **global** observability refers to the observability on the entire manifold where a state vector exists: if there is a single $\mathbf{x}_0$ for any output–input signals, then the system is called globally observable. The term **local** observability refers to the observability on some part of a manifold (submanifold) where the state vector exists: if there exist finitely many $\mathbf{x}_0$ for any output–input couple, then the system is locally observable. Otherwise, if there are infinitely many $\mathbf{x}_0$ for any output–input signals, the system is unobservable. In other words, observability analysis answers the question of whether two different initial conditions lead to two different outputs under the same inputs. There is no difference between local and global observability for linear time-invariant systems.

A necessary and sufficient condition for observability of linear systems, known as observability rank condition, states that if the observability matrix

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \in \mathbb{R}^{n_y n \times n_x} \quad (2)$$

has a full column rank ($\text{rank}(\mathcal{O}) = n_x$), then the linear system is observable. If the rank is not full, then the system is unobservable. Often $n = n_x$ is selected.³ However, since the rank of a square matrix is less than or equal to its number of columns, it is sufficient to consider only $n = \lceil \frac{n_x}{n_y} \rceil = \lfloor \frac{n_x}{n_y} \rfloor + 1$ row block matrices in the observability matrix, where $\lceil \cdot \rceil$ and $\lfloor \cdot \rfloor$ are, correspondingly, the ceiling and the floor operations. If $n_y = 1$, then $n = n_x$ and the observability matrix $\mathcal{O}$ is a square matrix.

² Kalman's original definition does not include the inputs (Kalman), while some other definitions include them (Leiva & Siegmund, 2003; Silverman & Meadows, 1967). For autonomous systems, inputs are not needed, but for general systems knowledge of inputs is required for state estimation. Therefore, for generality, we decided to include them.

³ Further in the paper it is assumed that $n = n_x$.

2.2. Global temporal observability of linear continuous systems

In practice, it is often desirable to evaluate the state vector over the entire interval when observations (output measurements) are taken. To accomplish this task, two steps are needed. First, the initial state vector $\mathbf{x}_0$ is evaluated from the output (observability analysis). Second, the obtained initial state vector, the state equation, and the known inputs are used to evaluate the state vector over the entire observation time window $0 \leq t \leq \tau$ using numerical integration. Even though numerical integration is not an issue for linear systems, it can be a challenging task for highly nonlinear and stiff dynamic systems. As a result, a single step that combines or replaces both steps may often be beneficial. To this end, we suggest a global temporal observability (GT-observability) concept.

Let us assume that each state vector component x_i , $i = 1, \dots, n_x$, is represented as a linear combination of n_c known (chosen) base functions⁴ $\phi_i(t)$, $i = 1, \dots, n_c$, as

$$x_i(t) = \sum_{j=1}^{n_c} z_{i,j} \phi_j(t), \quad (3)$$

where $z_{i,j}$, $i = 1, \dots, n_x$, $j = 1, \dots, n_c$, are unknown constants representing weights or control points. Base functions can be any set of orthogonal functions or spline basis functions. The base functions can be collected as a vector $\bar{\phi} = [\phi_1 \dots \phi_{n_c}]^T \in \mathbb{R}^{n_c}$ that spans a base (trial) space, here the superscript T means transposition. The vector of control points is defined similarly, $\mathbf{z} = [z_{1,1} \dots z_{1,n_c} \ z_{2,1} \dots z_{2,n_c} \dots z_{n_x,1} \dots z_{n_x,n_c}]^T \in \mathbb{R}^{n_z}$, where $n_z = n_x n_c$. In terms of control points, the state vector can be presented as

$$\mathbf{x}(t) = \Phi(t)\mathbf{z}, \quad (4)$$

where $\Phi(t) \in \mathbb{R}^{n_x \times n_x n_c}$ is the matrix of base functions

$$\Phi(t) = \begin{bmatrix} \bar{\phi}^T & 0 & \dots & 0 \\ 0 & \bar{\phi}^T & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \bar{\phi}^T \end{bmatrix}. \quad (5)$$

Since $\mathbf{z}$ is a constant vector, we have

$$\dot{\mathbf{x}}(t) = \dot{\Phi}(t)\mathbf{z}, \quad \ddot{\mathbf{x}}(t) = \ddot{\Phi}(t)\mathbf{z}, \quad \mathbf{x}^{(i)}(t) = \Phi^{(i)}(t)\mathbf{z}, \quad (6)$$

where $\mathbf{x}^{(i)}(t)$ is the i th order time derivative of $\mathbf{x}$. In essence, the state Eqs. (1) and the expressions (6) give two alternative descriptions for the time derivative of the state vector. If $\mathbf{z}$ is known, then the time evolution of the state vector is also known; and vice versa. Global temporal observability can be characterized as follows:

Definition 3. A system (1) is globally temporally observable (GT-observable) if, given an output $\mathbf{y}(t)$ and input $\mathbf{u}(t)$ signals over a finite time interval $0 \leq t \leq \tau$, it is possible to uniquely identify the vector of control points $\mathbf{z}$.

As its name suggests, global temporal observability focuses on finding the state vector function, which in turn provides state vector values at all observation times. The classical observability concept then becomes a **local temporal observability**; because it considers the possibility of finding the state vector at a specific moment in time (initial time). We warn the reader that the terms 'global' and 'local' now refer to two different aspects. One is related to *phase space* where the state vector exists, this leads to traditional global and local observability concepts (Hermann

& Krener, 1977; Sedoglavic, 2002). For instance, the system is globally (locally) observable if there is one (finite or countably infinite) initial condition(s) for a given input–output functions. Otherwise, the system is unobservable. Another aspect relates to *time*, hence the use of the word 'temporal'.

How can one establish global temporal observability? The simplest method is as follows. The system equations in terms of $\mathbf{z}$ is obtained if (4) and (6) are substituted into (1)

$$\begin{aligned} \dot{\Phi}\mathbf{z} &= A\Phi\mathbf{z} + B\mathbf{u}, \\ \mathbf{y} &= C\Phi\mathbf{z} + D\mathbf{u}. \end{aligned} \quad (7)$$

The observability matrix for the system (7) that we term the *global temporal observability matrix*, is defined as

$$\mathcal{O}_g = \begin{bmatrix} C\Phi \\ CA\Phi \\ CA^2\Phi \\ \dots \\ CA^{m-1}\Phi \end{bmatrix} \in \mathbb{R}^{n_y m \times n_z}. \quad (8)$$

We can select $m = n_z$. However, it is sufficient to consider only $m = \lceil \frac{n_z}{n_y} \rceil$ row block matrices in the observability matrix. One weakness of this method is that $\mathcal{O}_g$ requires $m - 2$ matrix multiplications of A with itself compared to $n - 2$ matrix multiplications in $\mathcal{O}$. Since $n_z = n_x n_c > n_x$, m in (8) is n_c times higher than n in (2). The reason for this is that the global temporal observability restores the state vector for the entire observable time duration, which requires more information.

A more convenient method for estimating global temporal observability is the following. Since basis functions are known, their time derivatives are also known. We can utilize this information to obtain an alternative form for the observability matrix (8). Let us define a row submatrix $\mathcal{P}$ of $\mathcal{O}_g$ as

$$\mathcal{P} = \begin{bmatrix} C\Phi \\ CA\Phi \\ CA^2\Phi \\ \dots \\ CA^{p-1}\Phi \end{bmatrix} \in \mathbb{R}^{n_y p \times n_z}, \quad (9)$$

where $p < m$. Then

$$\dot{\mathcal{P}} = \begin{bmatrix} C\dot{\Phi} \\ CA\dot{\Phi} \\ CA^2\dot{\Phi} \\ \dots \\ CA^{p-1}\dot{\Phi} \end{bmatrix} \in \mathbb{R}^{n_y p \times n_z} \quad (10)$$

and similarly for higher order time derivatives of $\mathcal{P}$. As a result,

$$\mathcal{O}_g = \begin{bmatrix} \mathcal{P} \\ \dot{\mathcal{P}} \\ \ddot{\mathcal{P}} \\ \dots \\ \mathcal{P}^{(r-1)} \end{bmatrix}, \quad (11)$$

where r satisfies the following identity $pr = m$ (more precisely, $r = \lceil \frac{n_z}{n_y p} \rceil$). For example, if $n_y = 1$ and $p = n_x$, then $m = n_z$ and $r = n_c$, so in this specific case $\mathcal{P}$ is similar to $\mathcal{O}$, with the only difference that Φ is present in $\mathcal{P}$. If $p = n_x$, then the block matrix multiplication, $\mathcal{P} = \mathcal{O}\Phi$, $\dot{\mathcal{P}} = \mathcal{O}\dot{\Phi}$, etc., gives a new expression

⁴ also called trial functions.

for

$$\mathcal{O}_g = \begin{bmatrix} \mathcal{O}\Phi \\ \mathcal{O}\dot{\Phi} \\ \mathcal{O}\ddot{\Phi} \\ \vdots \\ \mathcal{O}\Phi^{(n_c-1)} \\ \mathcal{O}\Phi \end{bmatrix} \in \mathbb{R}^{n_y n_z \times n_z}, \quad (12)$$

which requires only $n - 2$ matrix multiplications of A with itself (similar to $\mathcal{O}$). The observability matrix in (12) is estimated at a single point in time.

There is an alternative way to estimate the observability matrix $\mathcal{O}_g$. Instead of time differentiating $n_c - 1$ times Φ at a single point in time, n_c different points in time can be used without the need for time differentiation of Φ . Thus

$$\mathcal{O}_g = \begin{bmatrix} \mathcal{O}\Phi(t_1) \\ \mathcal{O}\Phi(t_2) \\ \mathcal{O}\Phi(t_3) \\ \vdots \\ \mathcal{O}\Phi(t_{n_c}) \end{bmatrix} \in \mathbb{R}^{n_y n_z \times n_z}. \quad (13)$$

The advantage of (13) is that it does not require time differentiation of Φ .

In fact, the increased number of variables n_z gives us more freedom to choose the data to estimate the unknown variable $\mathbf{z}$. For instance, $\mathcal{O}_g$ can also be estimated mixing two aforementioned ways,

$$\mathcal{O}_g = \begin{bmatrix} \mathcal{O}\Phi(t_1) \\ \mathcal{O}\dot{\Phi}(t_1) \\ \mathcal{O}\dot{\Phi}(t_2) \\ \mathcal{O}\Phi(t_3) \\ \vdots \\ \mathcal{O}\ddot{\Phi}(t_k) \end{bmatrix}. \quad (14)$$

The following two theorems characterize global temporal observability.

Theorem 1. *A necessary condition for a linear system (1) to be GT observable is that the observability matrix $\mathcal{O}_g$ has full column rank, i.e. $\text{rank}(\mathcal{O}_g) = n_z = n_x n_c$.*

Proof. The proof is a modification of the classical proof using observability matrix $\mathcal{O}$ (Wakin et al.). It is assumed that the input u and all its time derivatives are known. GT observability matrix (12) is used in the proof. Let us define a concatenated output vector $Y = [y^T \dot{y}^T \dots \overset{(n_x-1)}{y}^T]^T \in \mathbb{R}^{n_y n_x}$. Then using (7), we can write $Y = \mathcal{O}\Phi z + U$, where U is the vector that contains all the input terms. However, the last equation is an underdetermined linear system, so it cannot be utilized to find z yet. Let us define

a new concatenated vector $\bar{Y} = [Y^T \dot{Y}^T \dots \overset{(n_c-1)}{Y}^T]^T \in \mathbb{R}^{n_y n_x n_c}$, then $\bar{Y} = \mathcal{O}_g z + \bar{U}$, where $\bar{U}$ is the vector that contains all the input and its time derivative terms. The last matrix equation has a unique solution for z only if $\mathcal{O}_g$ has a full column rank. This completes the proof.

Theorem 2. *If a linear system is globally temporally observable, then it is locally temporally (classically) observable.*

Proof. The result is proved by contradiction. Let us assume that Theorem 2 is wrong, i.e. the system is GT observable, but classically unobservable. Both GT observability matrices (12) and

(13) can be used for a proof, but we will use the former, written as

$$\mathcal{O}_g = W\bar{\Phi}, \quad (15)$$

where the matrix $W \in \mathbb{R}^{n_y n_z \times n_z}$ is a block diagonal matrix composed of the classical observability matrix $\mathcal{O}$ as $W = \text{diag}[\underbrace{\mathcal{O} \dots \mathcal{O}}_{n_c}]$, while $\bar{\Phi} \in \mathbb{R}^{n_z \times n_z}$ is the matrix obtained from

Φ terms as $\bar{\Phi} = [\Phi^T \dots \overset{(n_c-1)}{\Phi}^T]^T$. The rank of the matrix W is equal to the sum of the ranks of its diagonal block matrices $\text{rank}(W) = \sum_i \text{rank}(\mathcal{O}_i) = n_c \text{rank}(\mathcal{O})$ (Meyer, 1970, 1973). Since the system is classically unobservable, $\text{rank}(W) = n_c \text{rank}(\mathcal{O}) < n_c n_x = n_z$. The rank of the corresponding GT observability matrix $\text{rank}(\mathcal{O}_g) = \min(\text{rank}(W), \text{rank}(\bar{\Phi})) < n_z$. However, for a GT-observability, it must be $\text{rank}(\mathcal{O}_g) = n_z$. Thus, we came to a contradiction, and the theorem is valid.

Therefore, for linear dynamic systems that satisfy the conditions for the existence and uniqueness of solutions, the two concepts are related: global temporal observability implies classical observability, but the converse statement is not true. For linear time-varying systems, the existence and uniqueness theorem is valid if A is continuous for $\mathbf{x}$ in some open set U in $\mathbb{R}^{n_x}$.

2.3. Classical observability of linear discrete systems

We briefly discuss the case of discrete dynamic systems considering the following linear time-invariant system:

$$\begin{aligned} \mathbf{x}(k+1) &= A_d \mathbf{x}(k) + B_d \mathbf{u}(k), \\ \mathbf{y}(k) &= C_d \mathbf{x}(k) + D_d \mathbf{u}(k), \end{aligned} \quad (16)$$

where $k \in \mathbb{Z}^{0+}$. By convention $\mathbf{x}(k) = \mathbf{x}(t_k)$. The observability is defined as follows

Definition 4. A system (16) is called observable if, given an output $\mathbf{y}(k)$ and input $\mathbf{u}(k)$ signals over a finite time interval $0 \leq k \leq N$, it is possible to uniquely identify the initial state vector $\mathbf{x}_0$.

The discrete linear system is observable if the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA_d \\ CA_d^2 \\ \vdots \\ CA_d^{n-1} \end{bmatrix} \in \mathbb{R}^{n_y n \times n_x} \quad (17)$$

has a full column rank. If the rank is not full, the system is unobservable. One often chooses $n = n_x$. The discretization process relates continuous and discrete state space representations as $A_d = e^{A\Delta t}$, $B_d = A^{-1}(A_d - I)B$, $C_d = C$, $D_d = D$, where Δt is the sampling time.

2.4. Global temporal observability of linear discrete systems

The extension of the global temporal observability concept to discrete systems is straightforward.

Definition 5. A system (16) is globally temporally observable (GT-observable) if, given an output $\mathbf{y}(k)$ and input $\mathbf{u}(k)$ signals over finite time interval $0 \leq k \leq N$, it is possible to uniquely identify the vector of control points $\mathbf{z}$.

Although the state vector is discrete, it is still possible to model it as a linear combination of known continuous base functions

$$x_i(k) = \sum_{j=1}^{n_c} z_{i,j} \phi_j(t)|_{t=t_k}. \quad (18)$$

Discretization of the state vector comes from the evaluation of the basis functions at integer values of the continuous independent variable dummy t . By defining the vectors $\tilde{\phi}$, $\mathbf{z}$, and the matrix of base functions as in (5), the state vector can be represented through a vector of control points as

$$\mathbf{x}(k) = \Phi(t)|_{t=t_k} \mathbf{z}. \quad (19)$$

As in the continuous case, there are two main ways to estimate the global temporal observability matrix. One way, similar to (12), is to use a single point in time as a reference point, but different orders of the finite time differences of the base function matrix

$$\mathcal{O}_g = \begin{bmatrix} \mathcal{O}\Phi \\ \mathcal{O}\Delta\Phi \\ \mathcal{O}\Delta^2\Phi \\ \vdots \\ \mathcal{O}\Delta^{(n_c-1)}\Phi \end{bmatrix}, \quad (20)$$

where $\Delta\Phi(k) = \Phi(k+1) - \Phi(k)$ and $\Delta^{(i)}\Phi(k) = \Phi^{(i-1)}(k+1) - \Phi^{(i-1)}(k)$, $i = 2, \dots, n_c - 1$. Another way, similar to (13), uses multiple points in time as reference points, but without any finite time differences of the base function matrix.

It is clear from (15) and the corresponding rank condition for $\mathcal{O}_g$ ($\text{rank}(\mathcal{O}_g) = \min(\text{rank}(W), \text{rank}(\tilde{\Phi}))$), that the system might be GT unobservable due to either unobservability (rank deficiency in $\mathcal{O}$) or poor choice of the basis functions (rank deficiency in $\tilde{\Phi}$). The following four problematic cases can be identified for the choice of basis functions.

- Case P1. For a continuous system, if any of the time derivatives of the basis functions up to order $n_c - 1$ become null, then the system becomes GT unobservable, see (12). Thus, the basis functions should be selected such that $\Phi^{(i)} \neq 0$ for all $1 \leq i < n_c$.
- Case P2. For a discrete system, if any of the finite time differences of the basis functions up to order $n_c - 1$ become null or less than the machine precision, then the system becomes GT unobservable, see (20). Thus, the basis functions should be selected such that $\Delta^{(i)}\Phi \neq 0$ for all $1 \leq i < n_c$.
- Case P3. For both continuous and discrete systems, if any of the two sampling times used to evaluate the basis functions give equal values, then the system becomes GT unobservable, see (13). Thus, the basis functions and time samples should be selected such that $\Phi(t_i) \neq \Phi(t_j)$ for any i, j such that $i \neq j$.
- Case P4. For both continuous and discrete systems, if a set of basis functions is not independent (orthogonal), then the system becomes GT unobservable.

Similar to Theorems 1 and 2 results are valid for discrete systems as well.

3. Simulations

3.1. Classical observability of continuous systems

To test the proposed global temporal observability concept, the following four continuous systems were considered:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, D = 0, \quad (21)$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = 0, \quad (22)$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}, D = 0, \quad (23)$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = 0. \quad (24)$$

For these systems $n_x = 3$, $n_y = 1$ for (21) and (23) or $n_y = 2$ for (22) and (24), $n_u = 1$. The input was chosen to be a sinusoidal function

$$u(t) = 30 \sin(10t). \quad (25)$$

The initial conditions were selected as $\mathbf{x}_0 = [1 \ 2 \ 3]^T$. The systems were simulated using the MATLAB's `lsim` command with the output recorded at a sampling time of 1 ms from 0 s to 3 s (resulting in $n_t = 3001$ time points). The rank of the observability matrix (2) was calculated to evaluate the observability of the systems.

For classically observable systems, in order to estimate the initial state vector $\mathbf{x}_0$, the vectors $Y \in \mathbb{R}^{n_y n_x}$ and $U \in \mathbb{R}^{n_y n_x}$ are introduced

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_{n_x} \end{bmatrix}, U = \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_{n_x} \end{bmatrix}. \quad (26)$$

The components of these vectors are evaluated at $t = 0$ as

$$Y_j = \begin{matrix} (j-1) \\ y \end{matrix}, U_j = \sum_{i=1}^{j-1} CA^{i-1}B \begin{matrix} (j-i-1) \\ u \end{matrix}, j = 1, \dots, n_x, \quad (27)$$

where $\begin{matrix} (0) \\ y \end{matrix} = y$ and $\begin{matrix} (0) \\ u \end{matrix} = u$. The initial state vector can be evaluated by solving the linear system of equations $\mathcal{O}\mathbf{x}_0 = (Y - U)$.

3.2. Global temporal observability of continuous systems

Four different families of basis functions were considered for the GT observability analysis: B-splines, trigonometric splines, polynomial ($\phi_j(t) = t^{j-1}$, $j = 1, \dots, n_c$), and exponential ($\phi_j(t) = e^{t(j-1)}$, $j = 1, \dots, n_c$). For each selected basis function, two ways to calculate the global temporal observability matrix $\mathcal{O}_g$ were considered: (12) evaluated at the initial time $t = 0$ and (13) evaluated at different equally spaced n_c time points within the 0-3 s interval. We chose $n_c = 4$, and therefore $n_z = 12$.

For calculating the vector of control points, the following procedure was implemented. To calculate $\mathcal{O}_g$ using (12), the vectors $\bar{Y} \in \mathbb{R}^{n_y n_z}$ and $\bar{U} \in \mathbb{R}^{n_y n_z}$ are introduced

$$\bar{Y} = \begin{bmatrix} \bar{Y}_1 \\ \bar{Y}_2 \\ \vdots \\ \bar{Y}_{n_z} \end{bmatrix}, \bar{U} = \begin{bmatrix} \bar{U}_1 \\ \bar{U}_2 \\ \vdots \\ \bar{U}_{n_z} \end{bmatrix}, \quad (28)$$

where

$$\bar{Y}_i = \begin{matrix} (j-1+k) \\ y \end{matrix}, \bar{U}_i = \sum_{l=1}^{j-1} CA^{l-1}B \begin{matrix} (j-1-l+k) \\ u \end{matrix}, i = 1, \dots, n_z. \quad (29)$$

Here the values of $k = 0, \dots, n_c - 1$ and $j = 1, \dots, n_x$ are found from $i = n_x k + j$, or in other words division of i with n_x gives the quotient k and the remainder j .⁵ The vectors (28) are evaluated at

⁵ The exception is that when the remainder is null or $j = 0$, then we take it as $j = n_x$ and subtract one from k .

$t = 0$. Alternatively one can calculate $\mathcal{O}_g$ using (13), the vectors $\bar{Y}$ and $\bar{U}$ are formed by stacking repeatedly the vectors Y and U from (26) evaluated at n_c different time points

$$\bar{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_{n_c} \end{bmatrix}, \quad \bar{U} = \begin{bmatrix} U_1 \\ U_2 \\ \dots \\ U_{n_c} \end{bmatrix}. \quad (30)$$

Finally, the vector of control points can be evaluated by solving the linear system of equations $\mathcal{O}_g \mathbf{z} = (\bar{Y} - \bar{U})$. To estimate the precision of the computation of the control points, the root mean squared error (RMSE) is utilized,

$$\varepsilon = \sqrt{\frac{\sum_{i=1}^{n_t} (\bar{y}(t_i) - y(t_i))^2}{n_t}}, \quad (31)$$

where $\bar{y}$ and $y = C\Phi \mathbf{z}$ are, respectively, the reference and the estimated outputs. If $\mathbf{z}$ is found, then the initial state vector $\mathbf{x}_0$ can be evaluated as $\mathbf{x}(0) = \Phi(0)\mathbf{z}$.

3.3. Classical observability of discrete systems

Four continuous systems (21)–(24) were discretized using the sampling time $\Delta t = 1$ ms. The rank of the observability matrix (17) was calculated to evaluate the observability of the systems. All steps and parameters are the same as for continuous systems, except that the components in vectors (26) are computed by using equations

$$Y_j = y(j), \quad U_j = \sum_{i=1}^{j-1} CA^{i-1}Bu(j-i), \quad j = 1, \dots, n_x. \quad (32)$$

3.4. Global temporal observability of discrete systems

Similar simulations were performed for discrete systems as for continuous systems. The only difference is that for GT observability matrix evaluated using (20), the components of the vectors $\bar{Y}$ and $\bar{U}$ in (28) are evaluated using

$$\bar{Y}_i = \Delta^{(k)}y(j), \quad \bar{U}_i = \sum_{l=1}^{j-1} CA^{l-1}B\Delta^{(k)}u(j-l). \quad (33)$$

Here $i = 1, \dots, n_z$, $k = 0, \dots, n_c - 1$, $j = 1, \dots, n_x$, and k and j are found from $i = n_x k + j$.⁶ Also $\Delta^{(0)}y(k) = y(k)$, $\Delta^{(0)}u(k) = u(k)$, and $\Delta^{(i)}y(k) = \Delta^{(i-1)}y(k+1) - \Delta^{(i-1)}y(k)$, $\Delta^{(i)}u(k) = \Delta^{(i-1)}u(k+1) - \Delta^{(i-1)}u(k)$, $i = 1, \dots, n_c - 1$.

4. Results

4.1. Continuous systems

The rank of the classical observability matrix $\mathcal{O}$ from (2) for each of the four systems are given in the vector $\text{rank} = [3 \ 3 \ 2 \ 2]$. Thus, the systems (21) and (22) are observable, while the other two systems are not. The results of the estimation of the initial state vector for two observable systems (21) and (22) are given in Table 1. The maximum relative error is around 7.5%.

The matrix rank analysis of $\mathcal{O}_g$ showed that $\text{rank} = [12 \ 12 \ 8 \ 8]$ for the four systems (21)–(24). These results indicate that the global temporal observability analysis leads to the conclusions similar to those in the classical observability analysis (the first two systems are globally temporally observable, while the remaining systems are not). The GT unobservability of systems (23)–(24) is due to one of the states being unobservable. These

Table 1

Estimated $\mathbf{x}_0$ for systems (21) and (22).

True	Continuous		Discrete	
	(21)	(22)	(21)	(22)
$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1.00 \\ 2.15 \\ 2.99 \end{bmatrix}$	$\begin{bmatrix} 1.00 \\ 2.14 \\ 3.00 \end{bmatrix}$	$\begin{bmatrix} 1.00 \\ 2.15 \\ 2.84 \end{bmatrix}$	$\begin{bmatrix} 1.00 \\ 2.15 \\ 3.00 \end{bmatrix}$

Table 2

Estimated ε for the systems (21)–(22). Four different basis functions are denoted as B for B-spline, T for trigonometric spline, P for polynomial, and E for exponential. M stands for method.

ε				
System (21)				
	Continuous		Discrete	
	M (12)	M (13)	M (20)	M (13)
B	592.73	93.91	N/A	93.91
T	585.90	$2.97 \cdot 10^3$	585.90	$2.97 \cdot 10^3$
P	592.73	93.91	N/A	93.91
E	$1.56 \cdot 10^5$	13.53	N/A	13.53
System (22)				
	Continuous		Discrete	
	M (12)	M (13)	M (20)	M (13)
B	$1.15 \cdot 10^3$ 5.72	2.87 0.40	989.36 6.52	3.01 0.39
T	148.14 24.51	267.99 83.72	91.84 24.52	268.78 83.68
P	$1.15 \cdot 10^3$ 5.72	2.87 0.40	989.36 6.52	3.01 0.39
E	$4.80 \cdot 10^5$ 126.52	6.90 0.01	$1.57 \cdot 10^5$ 0.31	7.62 0.00

rank results depend neither on the selected basis functions nor on the way of calculating $\mathcal{O}_g$. The RMSE results for the observable systems (21)–(22) are presented in Table 2. They show that the second method to estimate the GT observability matrix based on (13) is better than the first method using (12) or (20). This advantage is observed for both continuous and discrete systems and across all four families of basis functions. For the first system (21), the best performance is with the exponential basis functions, while for the second system (22), the B-spline and polynomial basis functions are the best. The results of estimation of $\mathbf{x}_0$ are $[1.00 \ 2.15 \ 2.99]^T$ for the system (21) and $[1.00 \ 2.14 \ 3.00]^T$ for the system (22), and these estimations do not depend on the chosen basis functions and the method of estimating the matrix $\mathcal{O}_g$. In addition, for investigating the influence of the choice of basis functions on the GT observability analysis, polynomial basis functions with two identical terms ($\phi_3(t) = \phi_2(t) = t$) were applied for two observable systems (21)–(22).⁷ In this case, the rank analysis shows the rank = 9, and it does not depend on the way $\mathcal{O}_g$ is calculated. As expected, the GT unobservability is due to the redundant spline basis function term (case P4). If redundant spline basis functions are used for unobservable systems, the rank becomes even smaller (rank = 6) because GT unobservability in this case is due to the combined effect of the unobservable state and redundant spline basis function terms. Thus, observable systems can be GT unobservable due to the wrong choice of basis functions.

⁶ The same conditions apply as in the previous footnote.

⁷ There is no point to apply them to unobservable systems.

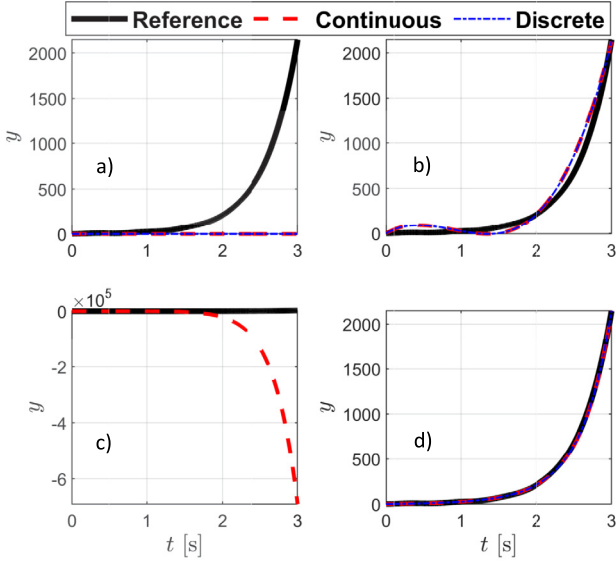


Fig. 1. Reference (black solid line) and estimated outputs of a continuous (red dotted) and discrete (blue dotted line) system (21). Upper subplots (a, b) are for B-spline, lower subplots (c, d) are for exponential basis functions. Left subplots (a, c) are for method (12) or (20), while right subplots (b, d) are for method (13). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Discrete systems

Due to the use of the same sampling times, the results for discrete systems are similar to the results for continuous systems. The estimated initial state vectors and the RMSE results for two observable systems (21) and (22) are also given in Tables 1 and 2, respectively. The only difference is that the GT observation test for the first system resulted in non-observation results for B-spline, polynomial, and exponential basis functions. The reason is that the finite difference of basis spline functions $\Delta^{(i)}\Phi(k)$ for degrees higher than 2 ($i \geq 3$) gives a number close to a machine precision value (case P2). As a result, the GT observability matrix becomes singular. If the sampling frequency is reduced, then the discrete system becomes GT observable. Thus, we conclude that a very short sampling time can affect the analysis for discrete systems. The results of estimation of $\mathbf{x}_0$ are $[1.00 \ 2.15 \ 2.83]^T$ for the system (21) and $[1.00 \ 2.15 \ 3.00]^T$ for the system (22), and these estimations do not depend on the choice of basis functions and the method for estimating $\mathcal{O}_g$. The reference and estimated outputs for continuous and discrete systems (21)–(22) are shown in Figs. 1–2, respectively. These figures show that the method of estimating the GT observability matrix using different points in the time domain (13) is superior to the methods using a single point in time domain (12) or (20). When basis functions give equal values at two different time samples (case P3), as expected, the systems become GT unobservable.

5. Discussion

Global temporal observability, taken as a *concept* or as an *idea*, is a generalization of the classical observability concept introduced by Kalman. This can be verified by assuming that $n_c = 1$ and $\phi_1 = 1$ for all x_i , see (3). Then $\Phi(t) = I$, where $I \in \mathbb{R}^{n_x \times n_x}$ is the identity matrix, and $\mathcal{O}_g = \mathcal{O}$, so GT observability reduces to the classical observability concept. However, GT observability, taken as a *property* of a system, is a stronger (stricter) property than classical observability. This is partly due to the rank requirement for the basis matrix Φ , which should be satisfied in addition

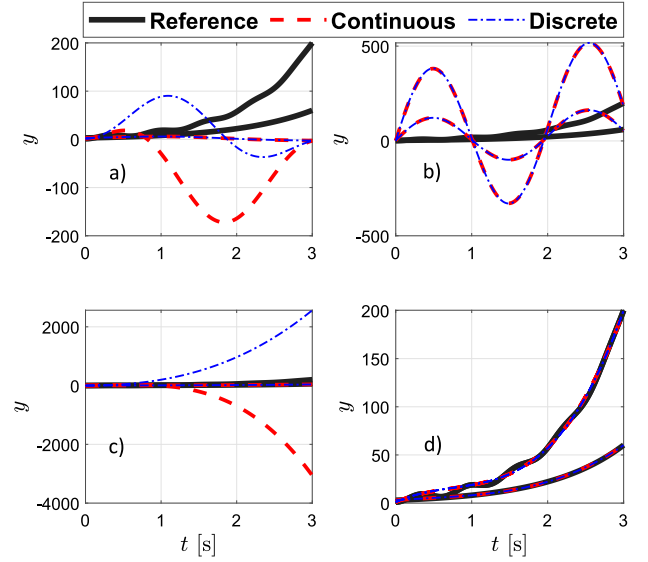


Fig. 2. Reference (black solid line) and estimated outputs of a continuous (red dotted) and discrete (blue dotted line) system (22). Upper subplots (a, b) are for trigonometric-spline, lower subplots (c, d) are for polynomial basis functions. Left subplots (a, c) are for method (12) or (20), while right subplots (b, d) are for method (13). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to the observability rank condition. Moreover, even if the system is GT observable, it does not guarantee that the restored states are correct. This is partly due to the dependence of the error (31) on the choice of the basis functions, see the results for trigonometric splines in Fig. 2. Therefore, Theorem 1 provides only the *necessary* condition, which itself is not sufficient. The simulation results indicate that the GT observability depends on the choice of basis functions (their number and degree) and the sampling time. It stands to reason then that the sufficient conditions for GTO are more difficult than the necessary conditions presented in this work. We can only speculate at this point in time that the sufficient conditions also require the investigation of numerical properties and other qualities of basis functions. We can only give vague or rough guidelines for the selection of basis functions: select basis functions as close in their behavior as possible to the ‘predicted’ or ‘estimated’ state behavior. For instance, if it is expected that a state function is a non-decreasing function of time, then it might be useful to select the basis function as polynomials. If, on the other hand, a state function is expected to be periodic, then trigonometric functions can be utilized as basis function. However, we leave this for future research.

There are two main differences between the classical observability matrix $\mathcal{O}$ and the global temporal observability matrix $\mathcal{O}_g$. The first one is the dimension of the matrices, with the latter matrix having more dimensions and a higher rank than the former. The second one is that for linear time-invariant systems, the observability matrix $\mathcal{O}$ is constant, while the global temporal observability matrix $\mathcal{O}_g$ is time-varying due to the presence of the time-dependent matrix of base functions Φ . For high-dimensional systems, the matrix inversion of GTO matrix is expected to be more computationally challenging than of the classical observability matrix. However, our method bypasses the integration step, which is required to get the state values for the entire time duration in the classical observability case. For high-dimensional systems, this time integration of state and observer dynamics might be more challenging than a single (albeit higher-dimensional) matrix inversion. Therefore, two factors

are at interplay: matrix inversion vs time integration. For high-dimensional linear systems, the sparsity of Φ matrix (5) might be utilized to reduce the computational cost of the matrix inversion operation, which in turn makes our approach more convenient. We again leave it for future research to determine which one is more efficient for high-dimensional systems.

From (27) it is clear that the maximum order of the time derivative of the output signal required for classical observability is $n_x - 1$. Similarly, from (29), the maximum order of the time derivative of the output signal required for GT observability is $n_x + n_c - 1$. This shows an advantage of evaluating $\mathcal{O}_g$ using (12) compared to using (8), which requires a time derivative of the output signal up to the order $n_x n_c - 1$, in case experimental data are utilized. In simulations, this advantage manifests itself in the fact that (12) requires a much smaller n_x number of state matrix multiplications with itself compared to (8) that requires $n_x n_c$ multiplications. Another advantage of using (12) is that it gives the constraint on the minimum degree of basis spline functions. For instance, the basis functions ϕ_i and their time derivatives up to order $n_c - 1$ should be nonzero. Otherwise, the global temporal observability matrix will not have the full column rank leading to the wrong conclusion that the system is not globally observable, even if it is in fact observable. As the number of control points per each state n_c is increased, the order of the time derivative also increases. This will have an undesirable effect on noisy measurements.

We note that from the practical point of view, there is no difference between the analysis of observability for continuous and discrete systems, as measurements are taken at discrete time samples in both cases. This means that the derivative is estimated using simple difference relations, and then the representation of the continuous system becomes equivalent to that of a discrete system. It was shown that the utilization of derivatives and differences for the observability matrix is equivalent (Aguirre & Letellier, 2005). It was also shown that the sampled data and derivatives are related by coordinate transformations (Gibson, Doyne Farmer, Casdagli, & Eubank, 1992). Our work is also related to state space reconstruction, where the task is to reconstruct the state space using its sampled data (Cao, Mees, & Judd, 1998; Gibson et al., 1992; Takens, 1981). The principal difference, however, is that we use output data instead of full-state data.

Global temporal observability depends on the choice of basis functions. As such, it mixes the observability problem with the curve-fitting problem, or, more specifically, with the residual method for solving differential equations. All problems associated with curve fitting also translate into problems of GT observability. Therefore, the global temporal observability property is embedded into the classical observability property. If a system is observable in the classical sense, then it might be respectively globally temporally observable (unobservable) if appropriate (inappropriate) basis functions are used. On the other hand, if a system is globally temporally observable, then it is observable in the classical sense. Let us contrast this with the functional observability property (Montanari et al., 2022): if a system is functionally observable, then it is not necessarily classically observable, while if a system is classically observable, then it is also functionally observable, see Fig. 3. The GT observability concept can be considered as 'offline' observability property based on measured time-data, while the Kalman filter or any other observers can be considered as an 'online' observers.

We assumed that all state vector components are described by the same basis functions. However, if each state vector component is discretized using different basis functions, then the only required modifications are in the matrix Φ :

$$\Phi(t) = \begin{bmatrix} \bar{\phi}_1 & 0 & \dots & 0 \\ 0 & \bar{\phi}_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \bar{\phi}_{n_x} \end{bmatrix}, \quad (34)$$

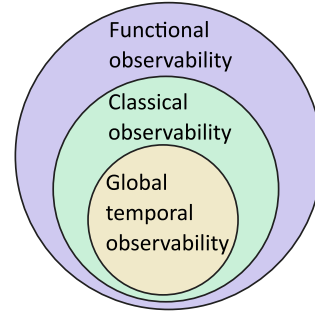


Fig. 3. The relationship between functional, classical, and global temporal observability.

where $\bar{\phi}_i = [\phi_{1,i} \dots \phi_{n_c,i}]^T \in \mathbb{R}^{n_c}$ is the vector of basis functions used to discretize state x_i , $i = 1, \dots, n_x$.

An important advantage of the proposed global temporal observability is that it restores the state vector for the whole time duration of interest in a single step, while the traditional approach requires two steps: initial state vector estimation and then integration. Thus, the time integration of dynamic equations is not necessary. However, one has to deal with an increased dimension of the observability matrix. Therefore, in each particular case, one has to decide what approach to observability is more efficient. If the dynamic system is such that numerical integration is problematic (stiff systems), but calculating the observability matrix is not, performing a global temporal observability analysis is advantageous. On the contrary, if finding the GT observability matrix presents problems, but the integration of a dynamic system is not, the traditional approach consisting of an observability analysis followed by time integration should be selected. Integration of dynamic equations for linear systems is not difficult. Therefore, we expect that the full potential of our method's advantages will be uncovered in our forthcoming research on observability of nonlinear systems. The representation of state vector as a linear combination of basis functions (4) is necessary to obtain the GTO matrix for the linear system. However, in the case of nonlinear systems, this assumption can be disregarded. The potential challenge of extending the GTO to nonlinear systems lies in the broadening of Lie differentiation operator to nonlinear state equations and basis functions. In effect, two Lie differentiations might be necessary: one with respect to the state equations followed by the second with respect to the basis functions.

6. Conclusions

In this paper, the concept of global temporal observability of linear dynamic systems is introduced. Its potential advantages and disadvantages are presented. The main advantage of the new concept is that it enables to restore the state information in a single step involving matrix inversion. It is shown that the global temporal observability concept is a generalization of the classical observability concept, but the global temporal observability property is a stronger property than the classical observability property. A new concept helps to establish a missing connection between state reconstruction, curve fitting, and system observability. Simulations were performed to investigate the properties of the proposed method. A necessary condition for global temporal observability is presented. The results demonstrate that global temporal observability depends on the selection of the basis functions.

CRedit authorship contribution statement

Altay Zhakatayev: Writing – original draft, Methodology, Investigation. **Yuriy Rogovchenko:** Writing – review & editing, Supervision, Conceptualization. **Matthias Pätzold:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors declare that no generative AI and/or AI-assisted technologies were used during the writing process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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